

Activity 1 — Data Science with Google Colab

Step 1 — Answer the class intake form first. Before any Colab work, answer the class intake & prior-knowledge form: <https://forms.gle/Z2AJpKTMx5CZcdFq7> (opens in a new tab). *Sagutan muna ito bago magsimula.* Everything below is **Step 2 onward**.



Fieldwork FIRST — Real Survey Fieldwork

The real work: you are **not** filling this out yourself. You go out and interview **REAL people, face to face**, and record their answers. This is what a data analyst actually does before any chart exists — they collect. *Hindi puwedeng kayo-kayo lang sa loob ng room. Lumabas kayo at mag-survey ng totoong tao.*

Do this BEFORE the Colab steps:

- Design a short survey (**4–6 questions**, mostly multiple choice) about a real topic in your community — commuting, food spending, phone/load use, study habits — anything you can ask strangers about.
- Collect at least **15 real respondents** — real people outside your own group (a terminal, a market, a canteen, your barangay, classmates from OTHER sections).
- Record each answer into your Google Form / Sheet as you go (or on paper then encode). Answers should come from **different people at different times** — not 15 rows typed in one minute.

PROOF (both required):

- A **PHOTO** of you actually surveying people in the field (you + a respondent, or you at the location with your form).
- A short **VIDEO (15–30 sec)** of you asking at least one real respondent your questions.

Shared rules: photo AND video are both required (no proof = not accepted) · pairs/small groups may survey together but each student needs their own 15+ and submits their own notebook · reduced/desk-based data only with the teacher's prior permission (message first) · your raw Sheet is a required deliverable, not just the charts.

Then: connect it in Colab

You'll connect **your own Google Form survey** to Google Colab and turn the **real responses you collected** into charts — bar charts, pie charts, a comparison between two questions, and a full survey dashboard built from data **you** gathered in the field.

Set up the form. Go to **forms.google.com** → create a form → Responses tab → green Sheets icon → **Create spreadsheet.** Use the **same Google account** that owns the form when you log into Colab — *magkaiba ang account, hindi gagana*. You'll copy the **Sheet ID** from the Sheet's URL (see Step 4).

Interpretation (write 3–4 sentences): who did you survey, where, and what is the ONE clearest pattern in your real data? What would you tell someone who could act on it?

Run each cell in Colab by clicking it and pressing **Shift + Enter**. Go in order, top to bottom.

Step 1 — Install the tools

This installs the libraries that let Colab talk to Google Sheets. Run it once.

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```
!pip install gspread google-auth
```

You should see: installation messages, ending with no errors (or "Requirement already satisfied").

Step 2 — Load the libraries

This brings in pandas (tables), matplotlib (charts), and the Google connection tools.

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```
import pandas as pd
import matplotlib.pyplot as plt
import gspread
from google.auth import default
from google.colab import auth

print("Setup complete!")
```

You should see: Setup complete!

Step 3 — Log in to Google

A popup will ask you to choose your Google account and allow access. Pick the account that **owns your form**, and click through every **Allow** / **Continue** screen.

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```
auth.authenticate_user()
creds, _ = default()
gc = gspread.authorize(creds)

print("Connected to Google!")
print("Ready to open your Google Sheet.")
```

Popup hiding? It sometimes appears behind the window or gets blocked. Check your address bar for a blocked-popup icon, or look behind the browser. *Sariling account nyo, hindi ang account ng trainer.*

You should see: Connected to Google!

Step 4 — Connect to YOUR sheet

Open your linked Google Sheet and look at the URL:

<https://docs.google.com/spreadsheets/d/1AbCdEf...1234567890/edit>

The **bold part between /d/ and /edit** is your Sheet ID. Paste it between the quotes below.

The ID below is only an example placeholder — it will fail until you replace it with **your own** Sheet ID.

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```
# ==== PASTE YOUR OWN SHEET ID BETWEEN THE QUOTES ====
SHEET_ID = "PASTE_YOUR_SHEET_ID_HERE"
# =====

# Connect and pull the data once
try:
    sheet = gc.open_by_key(SHEET_ID).get_worksheet(0)
    data = sheet.get_all_values()

    if len(data) > 1:
        df = pd.DataFrame(data[1:], columns=data[0])
        print(f"SUCCESS! Connected to your survey.")
        print(f"Responses found: {len(df)}")
        print(f"Questions (columns): {len(df.columns)}")
        print("\nYour questions:")
        for i, col in enumerate(df.columns):
            short = col[:60] + "..." if len(col) > 60 else col
            print(f"  Column {i}: {short}")
    else:
        print("Connected, but no responses yet.")
        print("Go fill out your form a few times, then run this cell again.")

except Exception as e:
    print(f"Connection failed: {e}")
    print("\nCheck these:")
    print("1. Is your SHEET_ID pasted correctly (no extra spaces)?")
    print("2. Are you logged in with the account that owns the form?")
    print("3. Did you run Step 3 first?")
```

You should see: SUCCESS!, the number of responses, and a numbered list of your questions. Remember those **Column numbers** — you'll use them to pick which question to chart.

Step 5 — See your full data table

Shows everything you collected as a clean table.

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```
if len(data) > 1:
    print(f"Rows (responses): {len(df)}")
    print(f"Columns (questions): {len(df.columns)}")
    print("\nYour complete data:")
    from IPython.display import display
    display(df)
else:
    print("No responses yet. Fill out your form first, then re-run Step 4.")
```

You should see: your full survey as a table.

Step 6 — Bar chart of one question

Counts how many people picked each answer. Change `question_column` to chart a different question (use the Column numbers from Step 4).

[COPY THIS CODE](#)

```

question_column = 1    # change this number to chart a different question

if len(data) > 1 and question_column < len(df.columns):
    question_name = df.columns[question_column]
    answer_counts = df.iloc[:, question_column].value_counts()

    print(f"Bar chart for: {question_name}")

    plt.figure(figsize=(12, 6))
    bars = plt.bar(answer_counts.index.astype(str), answer_counts.values)

    for bar, count in zip(bars, answer_counts.values):
        plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.05,
                 str(count), ha='center', va='bottom', fontweight='bold')

    plt.title(f'{question_name}', fontsize=14, fontweight='bold')
    plt.xlabel('Answers')
    plt.ylabel('Number of people')
    plt.xticks(rotation=45, ha='right')
    plt.grid(axis='y', alpha=0.3)
    plt.tight_layout()
    plt.show()

    print(f"Most popular answer: '{answer_counts.index[0]}' ({answer_counts.iloc[0]}
people)")
else:
    print("No data, or that question number does not exist. Check Step 4.")

```

You should see: a bar chart with counts on each bar. **Try it:** change the number to 2 and re-run — that one line is the whole trick.

Step 7 — Pie chart of one question

Same counts shown as percentages of the whole.

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```

question_column = 2    # change this number to chart a different question

if len(data) > 1 and question_column < len(df.columns):
    question_name = df.columns[question_column]
    answer_counts = df.iloc[:, question_column].value_counts()

    print(f"Pie chart for: {question_name}")

    plt.figure(figsize=(8, 6))
    plt.pie(answer_counts.values,
            labels=answer_counts.index.astype(str),
            autopct=lambda p: f'{p:.0f}%\n({int(round(p*answer_counts.sum()/100))})',
            startangle=90)
    plt.title(f'{question_name}', fontsize=14, fontweight='bold')
    plt.axis('equal')
    plt.tight_layout()
    plt.show()

    print(f"Most people answered: '{answer_counts.index[0]}'")
else:
    print("No data, or that question number does not exist. Check Step 4.")

```

You should see: a pie chart with percentages and counts.

Step 8 — Compare two questions (crosstab)

A crosstab shows how answers to one question relate to another, as a grouped bar chart.

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```

question_a = 1    # first question
question_b = 2    # second question

if len(data) > 1 and question_a < len(df.columns) and question_b < len(df.columns):
    name_a = df.columns[question_a]
    name_b = df.columns[question_b]

    # Build the crosstab: counts of each combination
    crosstab = pd.crosstab(df.iloc[:, question_a], df.iloc[:, question_b])

    print(f"How '{name_a}' relates to '{name_b}':\n")
    display(crosstab)

    crosstab.plot(kind='bar', figsize=(12, 6))
    plt.title(f'{name_a} vs {name_b}', fontsize=13, fontweight='bold')
    plt.xlabel(name_a)
    plt.ylabel('Number of people')
    plt.xticks(rotation=45, ha='right')
    plt.legend(title=name_b, bbox_to_anchor=(1.02, 1), loc='upper left')
    plt.grid(axis='y', alpha=0.3)
    plt.tight_layout()
    plt.show()

    print("Each group of bars is one answer to the first question.")
    print("The colors show how those people answered the second question.")
else:
    print("No data, or one of those question numbers does not exist. Check Step 4.")

```

You should see: a table and a grouped bar chart comparing two questions.

Step 9 — Full dashboard

Puts a pie chart for each question into one picture — a quick overview of your whole survey.

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```

if len(data) > 1:
    question_cols = list(range(1, len(df.columns))) # skip timestamp at column 0
    n = len(question_cols)

    cols = 3
    rows = (n + cols - 1) // cols
    fig, axes = plt.subplots(rows, cols, figsize=(6*cols, 5*rows))
    fig.suptitle('Survey Results Dashboard', fontsize=16, fontweight='bold')

    axes_flat = axes.flatten() if n > 1 else [axes]

    for i, qcol in enumerate(question_cols):
        counts = df.iloc[:, qcol].value_counts()
        axes_flat[i].pie(counts.values,
                        labels=counts.index.astype(str),
                        autopct='%1.0f%%',
                        startangle=90)

        title = df.columns[qcol]
        title = title[:30] + "..." if len(title) > 30 else title
        axes_flat[i].set_title(title, fontsize=10, fontweight='bold')

    # turn off any leftover empty boxes
    for j in range(n, len(axes_flat)):
        axes_flat[j].axis('off')

    plt.tight_layout()
    plt.show()
    print(f"Dashboard created for {n} questions.")
else:
    print("No data to build a dashboard yet. Fill out your form first.")

```

You should see: a dashboard of pie charts, one per question.

Tips & common errors

- Run cells **top to bottom** with Shift + Enter. If Step 4 fails, you probably skipped Step 3.
- **Connection failed?** Check the Sheet ID for typos/extra spaces, and confirm you logged in with the account that **owns the form**.
- **Empty charts?** You have no responses yet — fill out your form, then re-run Step 4.

- To chart a different question, just change the **question number** and re-run that cell.

What's next

In **Activity 2** you'll collect data a different way — by **scraping it off the web** with code.

Proof & Submit — required to unlock the next activity

Submit these on the activity page (the "Paste your link" box):

1. Your notebook / Colab link (the analysis).
2. Your RAW collected data — Sheet link, data file, or image-folder link.
3. Your PROOF: the required PHOTO(s) **and** the short VIDEO of your fieldwork.
4. Your written interpretation.

No fieldwork proof = the activity is NOT accepted, even if the code runs. *Kung talagang hindi kayo makakalabas para mag-collect, mag-message muna sa teacher BEFORE the activity para sa permission na gumamit ng reduced desk-based dataset — kung hindi, kailangan talaga ng fieldwork.*

Under the DPA / privacy notice: only collect data people agree to give; for photos/videos of people, ask their okay first; don't collect sensitive personal info you don't need.